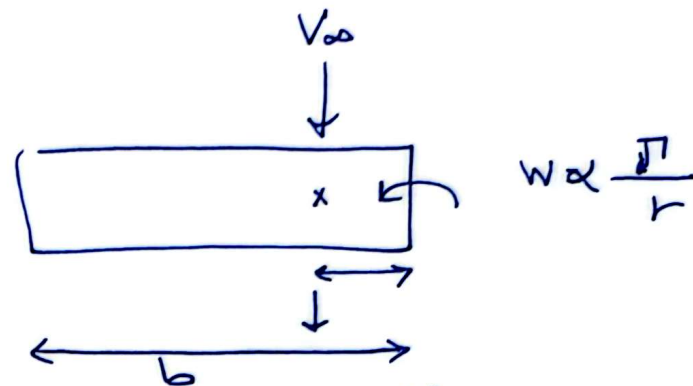


$$s = \frac{b}{2}$$



$$w \propto \frac{\Gamma}{r}$$

Downwash & Induced Drag

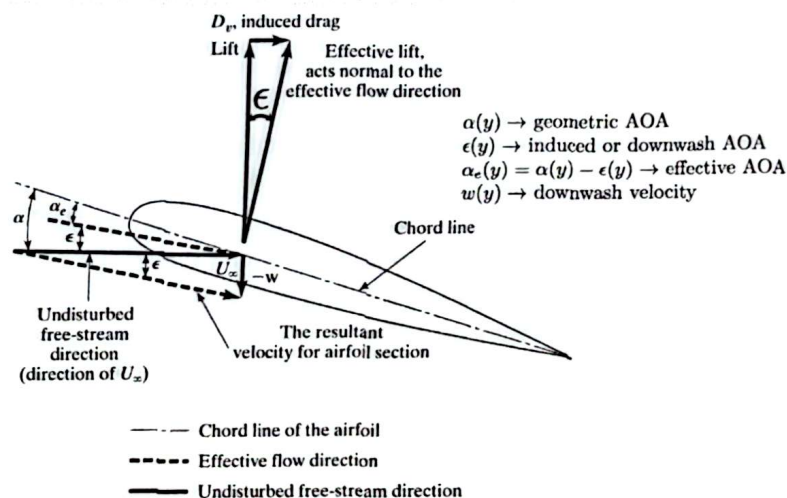
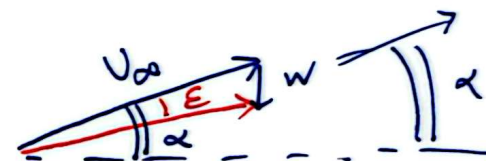
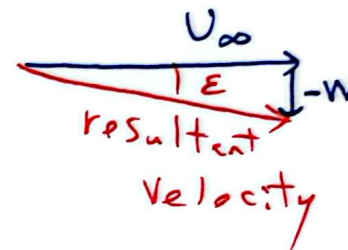


Figure 5.4: Downwash velocity and its effects, modified Fig. 7.6 from AE-BC

AE 319, EPG Pot. Wings, Qual. Effects, 202



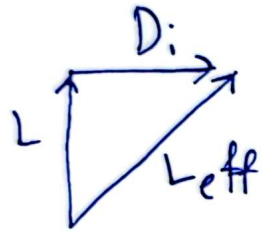
$$\alpha_{eff} = \alpha - \epsilon$$

Airfoil experience
 α_{eff} due to downwash!

w changes with location

$$w \propto \frac{\Gamma}{r}$$

$r =$. @ tip $\rightarrow w$ is high at tip
 $r = \frac{b}{2}$ @ root " " small " roots



D_i : induced drag is the negative effect of having vortices at the tip of the wing

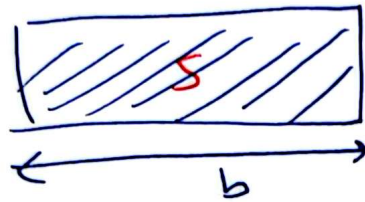
" D_i is called drag due to lift"

Downwash & Induced Drag

- ▶ Downwash $w(y)$ is a span-wise (i.e., across the wing span) velocity induced by the trailing edge vortices.
- ▶ Downwash is higher at the wing tips $y = \pm b/2$ than at the wing root $y = 0$. Hence it is a distribution $w(y)$
- ▶ This reduces the angle of attack α to $\alpha_e(y)$. (by an amount $\epsilon(y)$) This reduction is higher at the wing tip than at the root.
- ▶ This downwash $w(y)$ and the change in angle of attack also rotates the Lift vector by the angle ϵ which results in a force component in the free-stream direction known as the induced drag D_v for lift-induced drag. Note, $\epsilon(y)$ changes from the wing tip to the root.

Note, This is not an exact picture of what goes on. The actual flow is three-dimensional and more complicated.

C_L increases with $AR = \frac{b^2}{S}$ as the effect of wing tip vortices are reduced with increase of AR



Comparison aspect ratio

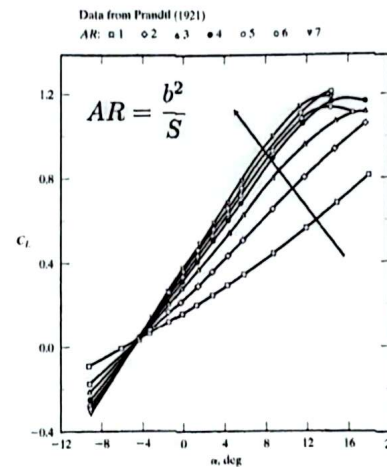
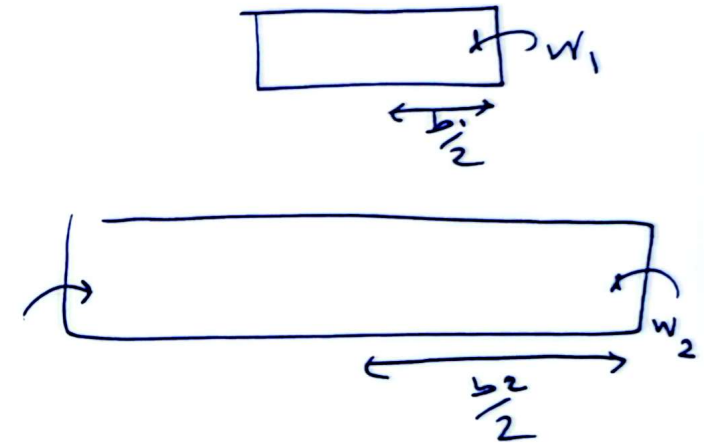


Figure 5.5: C_L vs. α aspect ratio effects, modified Fig. 7.9b from AE-BC

Note- These are from experiments and C_L is for a wing

AE 319, EPG Pot. Wings, Qual. Effects, 204



$$\frac{b_1}{2} < \frac{b_2}{2}$$

$$W \propto \frac{1}{r}$$

$$S_0 \quad W_1 > W_2$$

$$S_0 \quad D_{i_1} > D_{i_2}$$

$$E_{i_1} > E_{i_2}$$

$$\text{Airfoil} : C_{D_f} + C_{D_p}$$

$$\text{Wing} : C_{D_f} + C_{D_p} + \underbrace{C_{D_i}}_{= C_{D_v}}$$

Induced drag

$$\text{For potential flow} \begin{cases} C_{D_f} = 0 \\ C_{D_p} = 0 \end{cases} \text{ but } C_{D_i} \neq 0$$

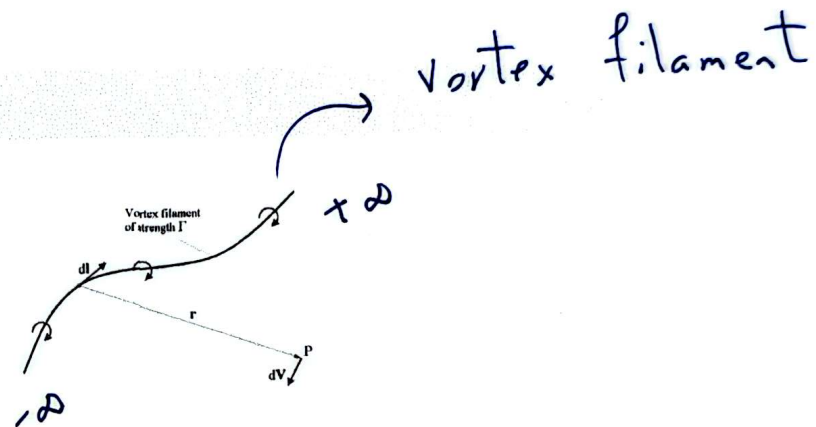
We approximate the drag on a wing by

$$C_D = \underbrace{C_{D2-D}}_{\text{form+skin friction drag}} + \underbrace{C_{D_v}}_{\text{induced drag}} \quad (5.1)$$

Note: for potential flows $C_{D2-D} \approx 0$

Biot-Savart Law

Image adapted from Wikipedia

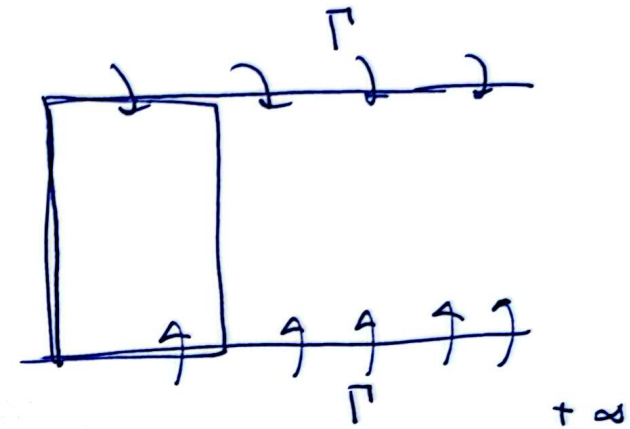


$$d\vec{V} = \frac{\Gamma}{4\pi} \frac{d\vec{l} \times \vec{r}}{|\vec{r}|^3} \quad (5.2)$$

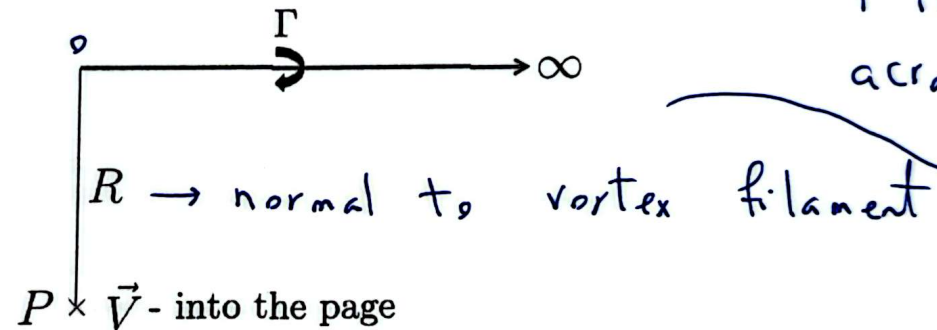
$$\text{Induced velocity} \rightarrow \boxed{\vec{V} = \frac{\Gamma}{4\pi} \int_C \frac{d\vec{l} \times \vec{r}}{|\vec{r}|^3}} \quad (5.3)$$

Velocity induced by vortex line, hence induced velocity.

Biot-Savart Law



Γ is constant across vortex filament

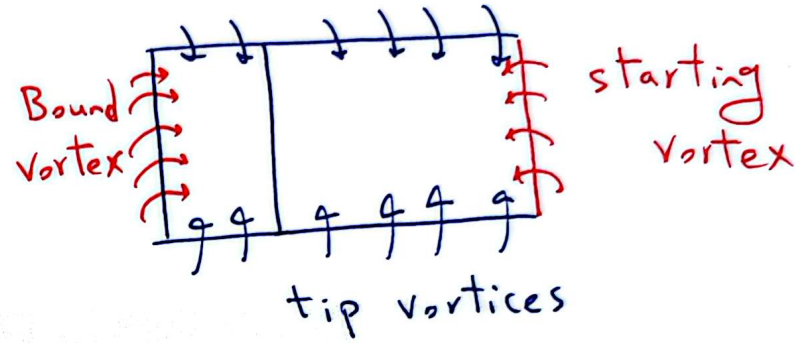
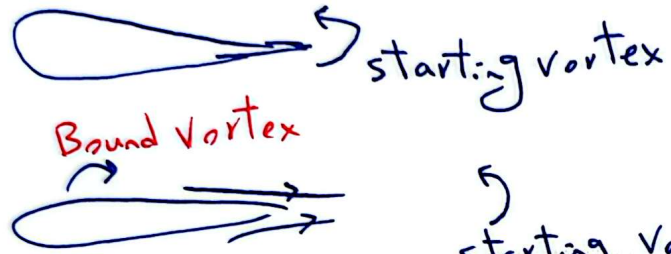


based on image from FA-JA

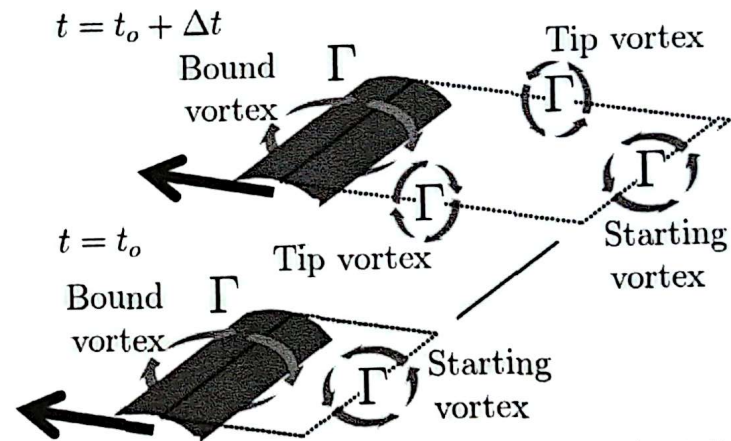
Velocity induced by vortex line extending to ∞ :

$$|\vec{V}| = \frac{\Gamma}{4\pi R}$$

(5.4)



starting vortex
Starting & bound vortex

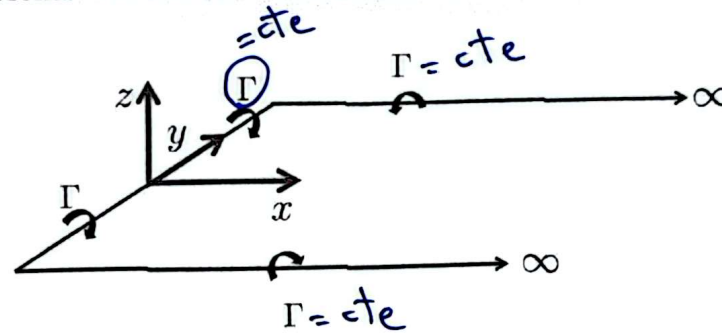


partially adapted from <http://www.oncmetrc.net/Design/Downwash/LiftLine/Line2.gif>

$$V = \frac{\Gamma}{4\pi R}$$

R : normal distance

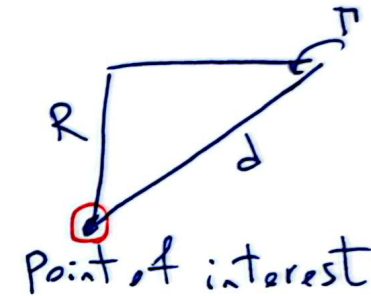
Problems with horse vortex model



single semi-infinite vortex $\rightarrow |\vec{V}| = \frac{\Gamma}{4\pi R}$ (5.5)

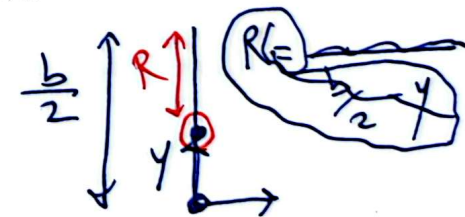
Therefore, downwash distribution $w_d(y)$ for horse shoe system:

$$w(y) = -\frac{\Gamma}{4\pi} \frac{b}{[(b/2)^2 - y^2]} \quad (5.6)$$



~~$V = \frac{\Gamma}{4\pi d}$~~

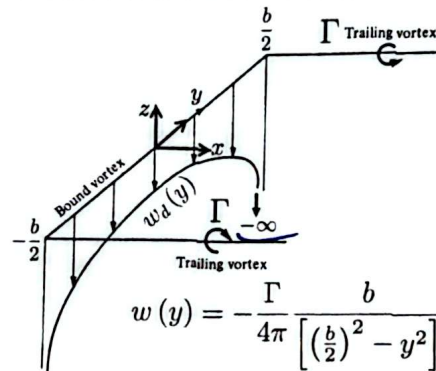
$$V = \frac{\Gamma}{4\pi R}$$



$$R = \frac{b}{2} - y$$

$$w \propto \frac{1}{\frac{b}{2} - y}$$

Problems with horse vortex model



partially adapted from FA-18

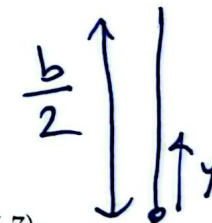
$$w\left(\pm \frac{b}{2}\right) = -\infty, \quad w(0) = -\frac{\Gamma}{\pi b} \quad \text{Horse shoe system - Not realistic} \quad (5.7)$$

(5.8)

$$w \propto \frac{1}{\frac{b}{2} - y}$$

when

$$y = \frac{b}{2}$$



$$w = \infty$$

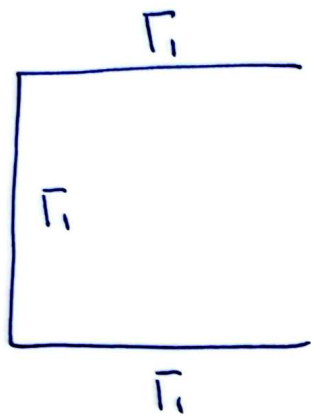
At the tip,

our model gives
infinity velocity

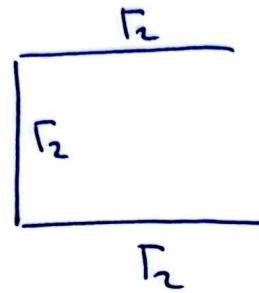
rational! ←

$$w(0) \propto \frac{1}{\frac{b}{2}}$$

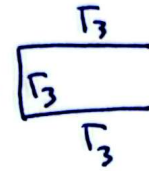
finite value



+

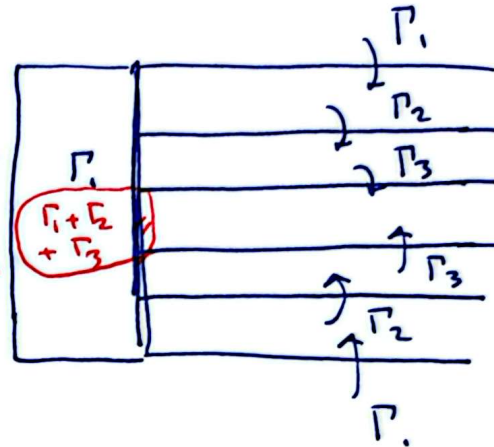


+

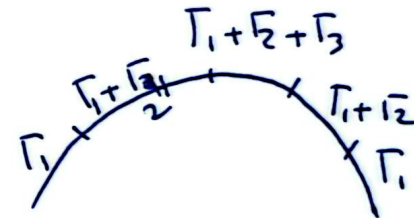
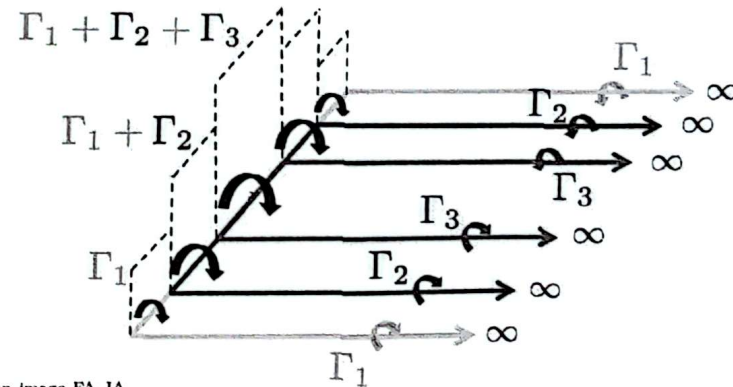


= $\Gamma_1 + \Gamma_2$

$\Gamma_1 + \Gamma_2$



Prandtl's lifting line theory



based on image FA-JA

We can superpose multiple horse-shoe vortex systems to get a more realistic representation. We can use superposition because we are considering potential flows.

$$dw_{y_1} = \frac{1}{4\pi} \frac{\left(\frac{d\Gamma}{dy} \times dy \right)}{y - y_1}$$

Vertex location
Point of interest

Prandtl's lifting line theory

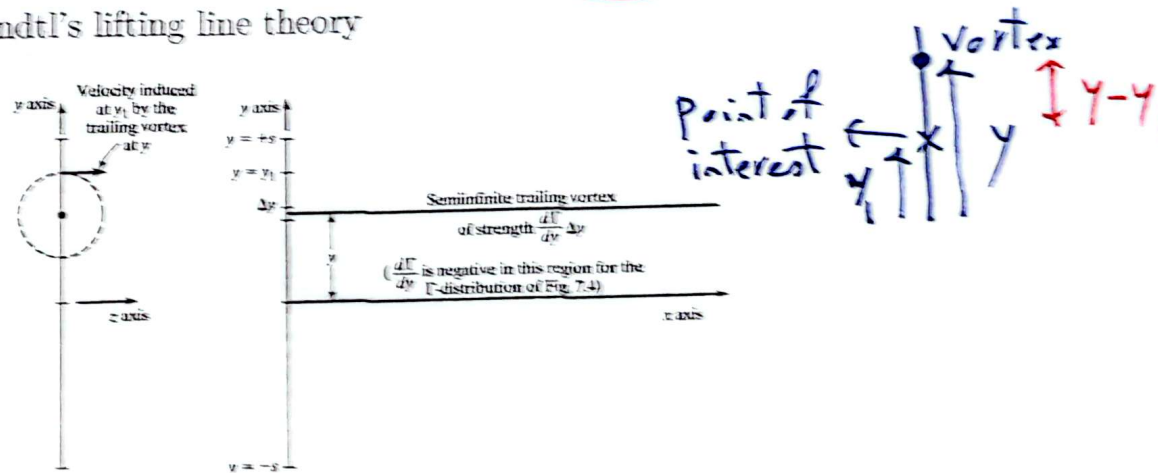


Figure 5.9: Induced velocity calculation, Fig. 7.5 from AE-BC

Downwash at y_1 due to vortex contribution from y

$$\delta w_{y1} = \frac{1}{2} \left[\frac{d\Gamma}{dy} dy \frac{1}{2\pi(y - y_1)} \right] \quad (5.12)$$

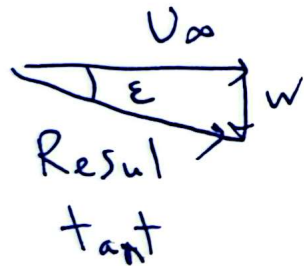
} vortices
x point of interest

Prandtl's lifting line theory

Therefore, the downwash w at y_1 due to the entire vortex sheet is,

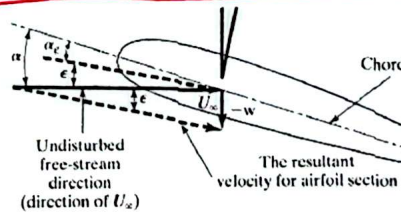
$$w(y_1) = \int_{-b/2}^{b/2} dw(y_0) = \frac{1}{4\pi} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy} \right) \frac{1}{(y - y_1)} dy \quad (\text{TEq. (7.2)}) \quad (5.13)$$

Downwash
velocity
by vortices



$$\epsilon = \frac{-w}{U_\infty}$$

$$\alpha_e = \alpha - \epsilon$$

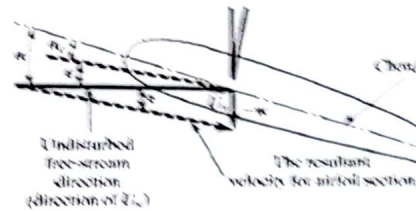


Using the figure, the induced or downwash AOA ϵ and the effective AOA α_e are

$$\tan \epsilon = \frac{-w}{U_\infty}, \text{ if } \epsilon \text{ is small } \tan \epsilon \approx \epsilon \approx -\frac{w}{U_\infty} \quad (\text{TEq. (7.3)}) \quad (5.14)$$

$$\alpha_e = \alpha - \epsilon \quad (\text{TEq. (7.4)}) \quad (5.15)$$

Prandtl's lifting line theory



Therefore substituting eq. (5.13) the induced AOA is

$$\epsilon = \frac{-w(y)}{U_\infty} = -\frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy} \right) \frac{dy}{(y-y_1)} \quad (5.16)$$

Downwash angle

$$\alpha_e = \alpha - \epsilon = \alpha + \frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy} \right) \frac{dy}{(y-y_1)} \quad (5.17)$$

$$\alpha = \alpha_e - \frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy} \right) \frac{dy}{(y-y_1)} \quad (5.18)$$

AE 514, ENG For. Wings, PLAT, 221

$\alpha = \alpha_e + w$
 Geometric angle of attack $\rightarrow$ effective A.O.A. $\rightarrow$ Downwash

$$C_l(y_1) = 2\pi(\alpha_e(y_1) - \alpha_{ol}(y_1))$$

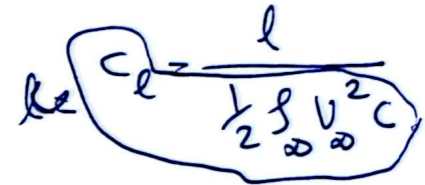
Prandtl's lifting line theory

From TAT (cambered airfoil:)

$$C_l(y_1) = 2\pi[\alpha_e(y_1) - \alpha_{ol}(y_1)] \quad (5.19)$$

$$l = \rho_\infty U_\infty \Gamma(y_1) = \frac{1}{2} C_l(y_1) U_\infty^2 c(y_1) \rho_\infty \quad (5.20)$$

$$C_l(y_1) = \frac{2\Gamma(y_1)}{U_\infty c(y_1)} \quad (5.21)$$



Sub eq. 5.21 in 5.19 5.21 5.19

$$\frac{2\Gamma(y_1)}{U_\infty c(y_1)} = 2\pi[\alpha_e(y_1) - \alpha_{ol}(y_1)] \quad (5.22)$$

$$\alpha_e(y_1) = \frac{\Gamma(y_1)}{\pi U_\infty c(y_1)} + \alpha_{ol}(y_1) \quad (5.23)$$

Prandtl's lifting line theory

Sub eq. 5.23 in 5.18

$$\alpha(y_1) = \underbrace{\frac{\Gamma(y_1)}{\pi U_\infty c(y_1)} + \alpha_{0l}(y_1)}_{\alpha_e} - \underbrace{\frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left[\frac{d\Gamma}{dy} \right] \frac{dy}{(y - y_1)}}_{\epsilon} \quad (5.24)$$

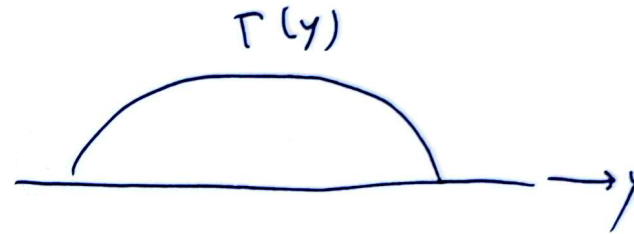
Fundamental Eqn. of Prandtl's lifting line theory:

$$\alpha(y_1) = \frac{\Gamma(y_1)}{\pi U_\infty c(y_1)} + \alpha_{0l}(y_1) + \underbrace{\frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left[\frac{d\Gamma}{dy} \right] \frac{dy}{(y_1 - y)}}_{\epsilon} \quad (5.25)$$

"changed
the denominator"

Solve for $\Gamma(y)$, Note y is the same as y_1 i.e., y_1 is a dummy variable

Prandtl's lifting line theory



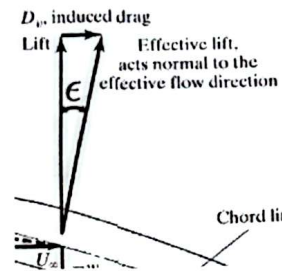
Once $\Gamma(y)$ is known:

1. $l(y) = \rho_{\infty} U_{\infty} \Gamma(y) \rightarrow \text{Load distribution}$ (5.26)

2. $L = \int_{-b/2}^{b/2} l(y) dy = \int_{-b/2}^{b/2} \rho_{\infty} U_{\infty} \Gamma(y) dy$ (TEq. (7.6)) (5.27)

$$C_L = \frac{L}{q_{\infty} S} = \frac{2}{U_{\infty} S} \int_{-b/2}^{b/2} \Gamma(y) dy \quad (5.28)$$

Prandtl's lifting line theory



$$\tan \epsilon = \frac{d_i}{l}$$

$$\rightarrow d_i = l \tan \epsilon \rightarrow d_i \approx l \epsilon$$

$$3. d_v \approx l \epsilon \text{ (for small } \epsilon) \rightarrow d_v(y) \approx l(y) \epsilon(y)$$

$$= \rho_\infty U_\infty \Gamma(y) \left(\frac{-w(y)}{U_\infty} \right) = -\rho_\infty \Gamma(y) w(y) \quad (5.29)$$

$$D_v = \int_{-b/2}^{b/2} d_v dy = - \int_{-b/2}^{b/2} \rho_\infty \Gamma(y) w(y) dy \quad (\text{TEq. (7.7)}) \quad (5.30)$$

$$C_{Dv} = \frac{D_v}{q_\infty S} = \frac{2}{U_\infty^2 S} \int_{-b/2}^{b/2} \Gamma(y) w(y) dy \quad (5.31)$$

Elliptical lift distribution

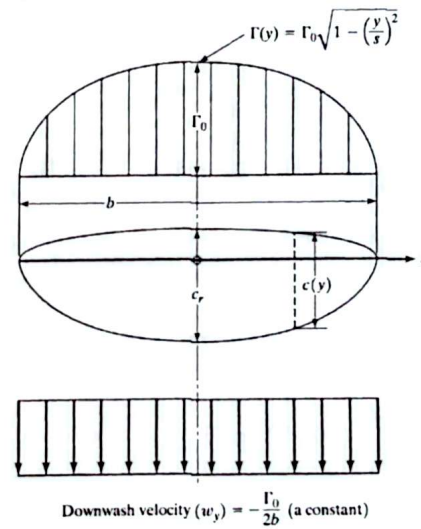
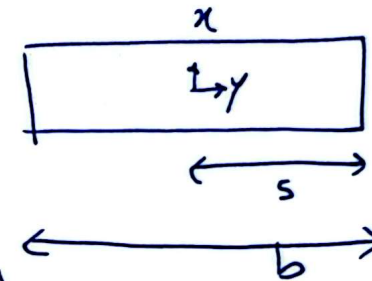


Figure 5.10: Elliptical lift distribution, Fig. 7.5 from AE-BC



Assume:

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{y}{s}\right)^2}$$

$$\Gamma(0) = \Gamma_0$$

$$\Gamma(s) = 0$$

$$s = \frac{b}{2}$$

$$\Gamma = \Gamma_0 \sqrt{1 - \left(\frac{y}{s}\right)^2} = \Gamma_0 \sqrt{1 - \left(\frac{y}{b/2}\right)^2}$$

Elliptical lift distribution

Elliptical load distribution:

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} \quad (5.32)$$

$y = 0$ @ root, $\Gamma(y=0) = \Gamma_0$, $y = \pm \frac{b}{2} = \pm s$ @ span $\Gamma(y = \pm \frac{b}{2} = \pm s) = 0$

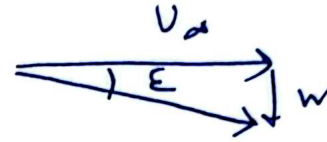
Root tip

$$w(y_1) = \frac{1}{4\pi} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy} \right) \frac{1}{4\pi(y-y_1)} dy \quad (\text{TEq. (7.2)})$$

After "some" algebra

$$w_d(y_0) = -\frac{\Gamma_0}{2b} = -\frac{\Gamma_0}{4s} \quad (\text{TEq. (7.10)}) \quad (5.33)$$

Therefore, the downwash distribution for an elliptically loaded wing is constant



$$\epsilon = \tan^{-1} \frac{-w}{U_\infty} \approx \frac{-w}{U_\infty}$$

Elliptical lift distribution

$$\epsilon = -\frac{w_d}{U_\infty} = \frac{\Gamma_0}{2bU_\infty} = \frac{\Gamma_0}{4sU_\infty} \quad (1) \quad (5.34)$$

Therefore, since the downwash distribution for an elliptically loaded wing is constant the induced angle of attack is also a constant

$$\begin{cases} S : \text{area} \\ s = \frac{b}{2} \end{cases}$$

$$C_L = \frac{L}{q_\infty S} = \frac{2}{U_\infty S} \int_{-b/2}^{b/2} \Gamma(y) dy \quad (5.35)$$

$$= \frac{2}{U_\infty S} \int_{-b/2}^{b/2} \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} dy \quad (5.36)$$

$$= \frac{2}{U_\infty S} \frac{\Gamma_0 b \pi}{4} = \frac{\pi b^2}{S} \underbrace{\frac{\Gamma_0}{2bU_\infty}}_{\epsilon} = \frac{\pi \epsilon b^2}{S} \quad (\text{TEq. (7.12)}) \quad (5.37)$$

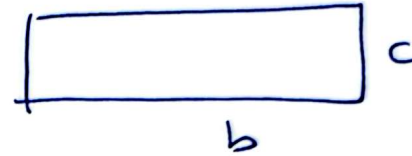
$$C_L = \frac{\pi \epsilon b^2}{S}$$

lift coeff. for elliptical Γ distribution

$$AR = \frac{b^2}{S}$$

$$\Rightarrow C_L = \frac{\pi \epsilon AR}{AR}$$

$$\Rightarrow \epsilon = \frac{C_L}{\pi AR}$$



$$AR = \frac{b^2}{S} = \frac{b^2}{\underbrace{bc}_{\text{rectangular wing}}} = \frac{b}{c}$$

Elliptical lift distribution

Aspect ratio:

$$AR = \frac{b}{c} = \frac{b^2}{bc} = \frac{b^2}{S} \quad (5.38)$$

Therefore,

$$\epsilon = \frac{C_L}{\pi AR} \text{ for elliptical loading (TEq. (7.14))} \quad (5.39)$$

$$\underline{\underline{C_{Dv} = \frac{2}{U_\infty S} \int_{-b/2}^{b/2} \Gamma(y) \underbrace{\epsilon(y)}_{\text{const.}} dy = C_L \epsilon}} \quad (5.40)$$

$$C_{Dv} = \frac{C_L^2}{\pi AR}, \quad AR \rightarrow \infty, \quad C_{Dv} \rightarrow 0 \quad (2-D \text{ wing/airfoil}) \quad (\text{TEq. (7.17)}) \quad (5.41)$$

$$\begin{aligned} C_{Di} &= C_L \epsilon \\ &= C_L \frac{C_L}{\pi AR} \\ \underline{\underline{C_{Di} &= \frac{C_L^2}{\pi AR}}} \end{aligned}$$

$$\text{elliptic} \rightarrow \Gamma = \Gamma_0 \sqrt{1 - \left(\frac{y}{\frac{b}{2}}\right)^2} = \Gamma_0 \sin \theta$$

Generalized $\rightarrow$

$$\Gamma = 2bU_\infty \sum_{n=1}^{\infty} A_n \sin(n\theta)$$

$$\text{Let } y = -\frac{b}{2} \cos \theta = -s \cos \theta, \quad y_1 = -\frac{b}{2} \cos \theta_1 \quad (5.42)$$

$$\text{and } \Gamma(y) = 2bU_\infty \sum_{n=1}^{\infty} A_n \sin(n\theta) \rightarrow \text{Fourier sine series (TEq. (7.21))} \quad (5.43)$$

Note: Elliptic load dist.

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} \quad (5.44)$$

$$\Gamma(\theta) = \Gamma_0 \sqrt{1 - \cos^2 \theta} = \Gamma_0 \sin \theta \rightarrow A_1 = \frac{\Gamma_0}{2bU_\infty}, \rightarrow A_n = 0 \text{ (for } n \neq 1) \quad (5.45)$$

$$\frac{y}{\frac{b}{2}} = -\cos \theta \quad \text{or} \quad \frac{y}{s} = -\cos \theta$$

$$\frac{y_1}{s} = -\cos \theta_1$$

Generalized lift distribution

We enforce the equation at N spanwise locations

$$\theta_{1n} = \frac{n\pi}{N+1}, \quad n = 1, 2, \dots, N$$

We avoid $\theta_1 = 0$ and π as $\sin \theta_1 = 0$ there. We then solve the resulting N linear equations for the N unknowns. Once A_n 's are calculated then,

$$C_L = \frac{2}{U_\infty S} \int_{-b/2}^{b/2} \Gamma(y) dy = \frac{2b^2}{S} \sum_{n=1}^N A_n \underbrace{\int_0^\pi \sin(n\theta) \sin \theta d\theta}_{\begin{cases} = 0, & n \neq 1 \\ = \frac{\pi}{2}, & n = 1 \end{cases}} \quad (5.48)$$

$$C_L = A_1 \pi \frac{b^2}{S} = A_1 \pi AR \quad (\text{TEq. (7.27)}) \quad (5.49)$$

for generalized lift distribution

$$C_L = A_1 \pi \frac{b^2}{S} = \boxed{A_1 \pi AR}$$

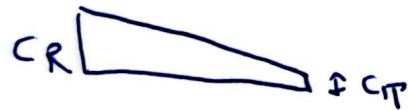
"the higher AR $\rightarrow$ the higher C_L "

Generalized lift distribution

Therefore,

$$C_{Dv} = \pi \frac{b^2}{S} \sum_{n=1}^N n A_n^2 = \underbrace{\pi AR A_1^2}_{=C_L^2/(\pi AR)} \left[1 + \sum_{n=2}^N n \left(\frac{A_n}{A_1} \right)^2 \right] \quad (5.54)$$

$$= \underbrace{\frac{C_L^2}{\pi AR}}_{C_{Di-ellip}} (1 + \delta) \quad \text{where } \delta = \sum_{n=2}^N n \left(\frac{A_n}{A_1} \right)^2 \geq 0 \quad (5.55)$$



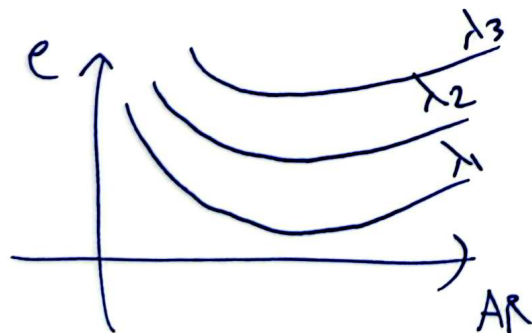
Let $e = \frac{1}{1+\delta}$, then (5.56)

$$C_{Dv} = \frac{C_L^2}{\pi e AR} \quad (\text{TEq. (7.29)}) \quad (5.57)$$

$e \rightarrow$ span efficiency factor, $e = 1$ for elliptical loading and $e < 1$ for any other loading.

$$\lambda = \frac{C_T}{C_R}$$

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For general lift distribution

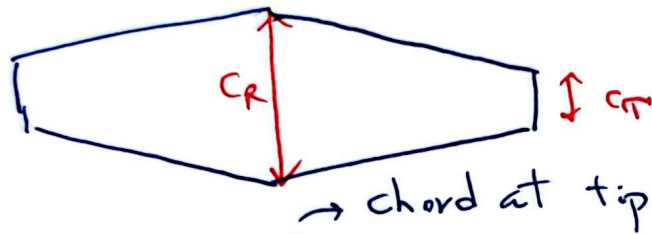
$$C_{Di} = \frac{C_L^2}{\pi AR} (1 + \delta) = \frac{C_L^2}{\pi AR e}$$

$$\delta = \sum_{n=2}^N n \left(\frac{A_n}{A_1} \right)^2$$

wing efficiency factor

Realistic model

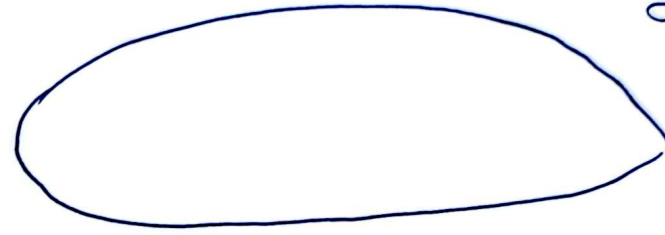
Best induced drag



$$\lambda = \frac{c_T}{c_R}$$

chord at root

Elliptical lift distribution



"elliptical wing has elliptical lift distribution"

1. Choose $c(y_1)$ = elliptical elliptically shaped planform , $\alpha(y_1)$ = constant, $\alpha_{0l}(y_1)$ = constant.
2. Choose $c(y_1)$ = constant rectangular planform , $\alpha(y_1)$ = elliptical - geometric twist , $\alpha_{0l}(y_1)$ = constant.
3. Choose $c(y_1)$ = constant rectangular planform , $\alpha(y_1)$ = constant, $\alpha_{0l}(y_1)$ = elliptical - aerodynamic twist .
4. Combination of above.

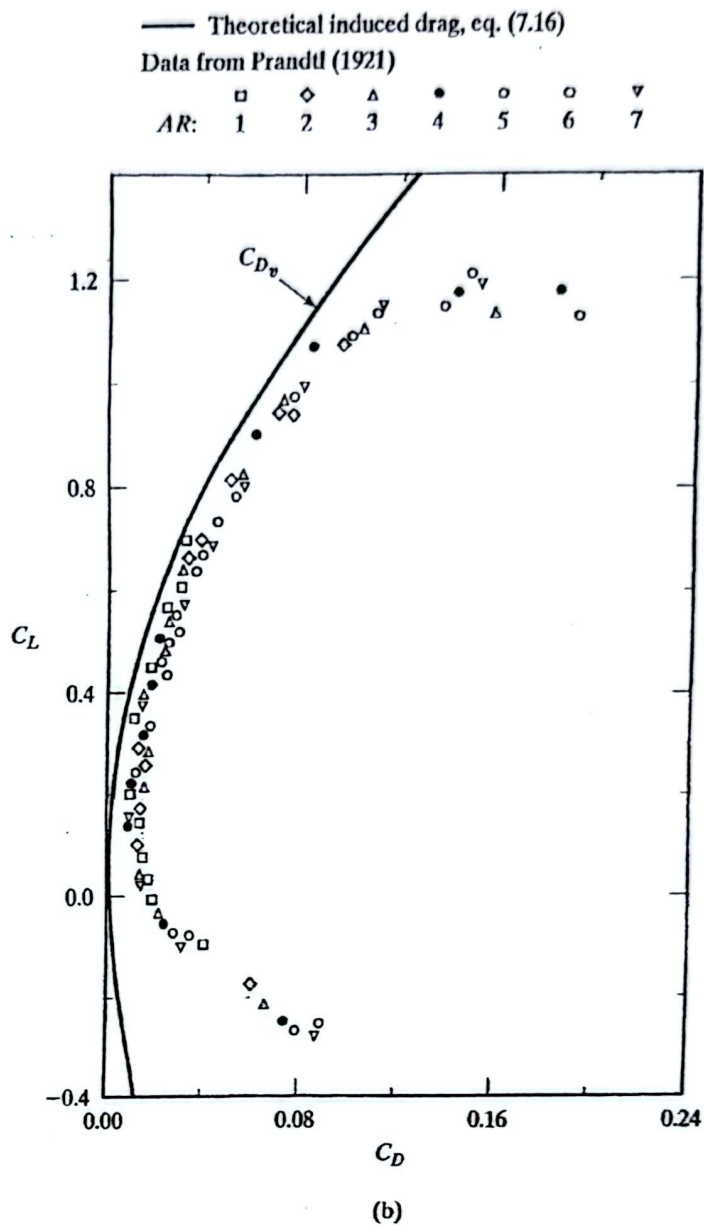


Figure 7.9 (continued) (b) drag polars converted to $AR = 5$.

As was done previously, the physical spanwise coordinate (y) has been replaced by the ϕ coordinate:

$$\frac{y}{s} = -\cos \phi$$

A sketch of one such Fourier series is presented in Fig. 7.11. Since the spanwise lift distribution represented by the circulation of Fig. 7.11 is symmetrical, only the odd terms remain.

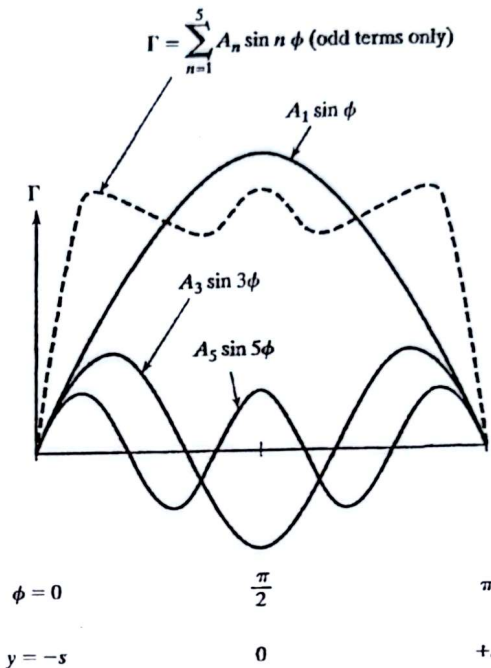


Figure 7.11 Symmetric spanwise lift distribution as represented by a sine series.

Let the equivalent lift-curve slope $(dC_l/d\alpha)_0$ be designated by the symbol a_0 . Note that since $\alpha_e = \alpha - \epsilon$, equations (7.23) and (7.24) can be combined to yield the relation

$$\frac{2\Gamma(\phi)}{U_\infty c} = a_0 (\alpha - \epsilon) \quad (7.25)$$

$$\frac{2\Gamma(\phi)}{c(\phi)a_0} = U_\infty [\alpha(\phi) - \alpha_{0l}(\phi)] - U_\infty \epsilon(\phi)$$

For the present analysis, five parameters in equation (7.25) may depend on the spanwise location ϕ (or, equivalently, y) at which we will evaluate the terms. The five parameters are (1) Γ , the local circulation; (2) ϵ , the downwash angle, which depends on the circulation distribution; (3) c , the chord length, which varies with ϕ for a tapered wing planform; (4) α , the local geometric angle of attack, which varies with ϕ when the wing is twisted (i.e., geometric twist, which is illustrated in Fig. 5.7), and (5) α_{0l} , the zero lift angle of attack, which varies with ϕ when the airfoil section varies in the spanwise direction (which is known as *aerodynamic twist*). Note that

$$U_\infty \epsilon = -w = -\frac{1}{4\pi} \int_{-s}^{+s} \frac{d\Gamma/dy}{y - y_1} dy$$

Using the Fourier series representation for Γ and the coordinate transformation, we obtain

$$-w = U_\infty \frac{\sum n A_n \sin n\phi}{\sin \phi}$$

Downwash for generalized lift dist.

$$\frac{2\Gamma(\phi)}{U_\infty c} = a_0 (\alpha - \epsilon)$$

$$\frac{2\Gamma(\phi)}{a_0 c(\phi)} = U_\infty (\alpha - \epsilon)$$

$$U_\infty (\alpha - \epsilon) = \frac{2\Gamma(\phi)}{a_0 c(\phi)}$$

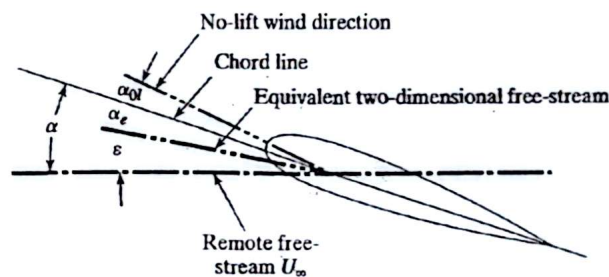
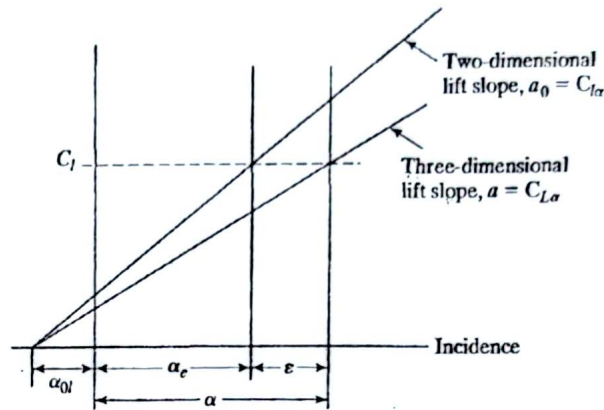
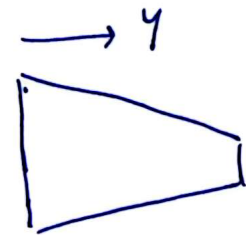


Figure 7.12 Nomenclature for wing/airfoil lift.



$$C = C(\gamma)$$

Equation (7.25) can be rewritten as

$$\frac{2\Gamma}{ca_0} = U_\infty(\alpha - \alpha_{0l}) - U_\infty \frac{\sum n A_n \sin n\phi}{\sin \phi}$$

Since $\Gamma = 4sU_\infty \sum A_n \sin n\phi$, the equation becomes

$$\frac{8s}{ca_0} \sum A_n \sin n\phi = (\alpha - \alpha_{0l}) - \frac{\sum n A_n \sin n\phi}{\sin \phi}$$

Defining $\mu = ca_0/8s$, the resultant governing equation is

$$\mu(\alpha - \alpha_{0l}) \sin \phi = \sum A_n \sin n\phi (\mu n + \sin \phi) \quad (7.26)$$

which is known as the **monoplane equation**. If we consider only symmetrical loading distributions, only the odd terms of the series need be considered. That is, as shown in the sketch of Fig. 7.11,

$$\Gamma(\phi) = 4sU_\infty(A_1 \sin \phi + A_3 \sin 3\phi + A_5 \sin 5\phi + \dots)$$

$$a_0: c_l (\text{airfoil})$$

$$s = \frac{b}{2}$$